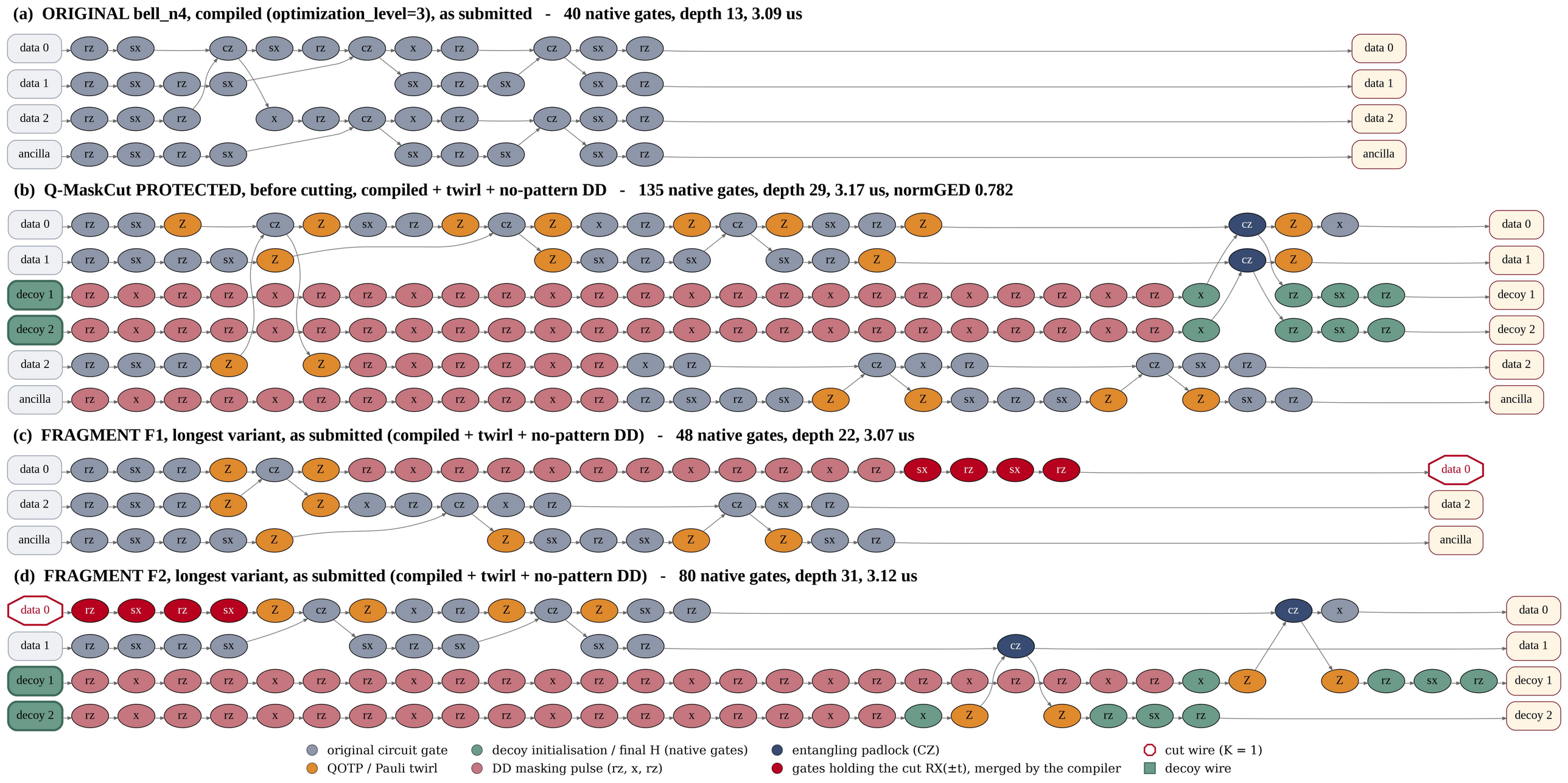

(a) ORIGINAL bell_n4, compiled (optimization_level=3), as submitted - 40 native gates, depth 13, 3.09 us
(b) Q-MaskCut PROTECTED, before cutting, compiled + twirl + no-pattern DD - 135 native gates, depth 29, 3.17 us, normGED 0.782
(c) FRAGMENT F1, longest variant, as submitted (compiled + twirl + no-pattern DD) - 48 native gates, depth 22, 3.07 us
(d) FRAGMENT F2, longest variant, as submitted (compiled + twirl + no-pattern DD) - 80 native gates, depth 31, 3.12 us
data 0
data 1
data 2
ancilla
decoy 1
decoy 2
original circuit gate
QOTP / Pauli twirl
decoy initialisation / final H (native gates)
DD masking pulse (rz, x, rz)
entangling padlock (CZ)
gates holding the cut RX(±t), merged by the compiler
cut wire (K = 1)
decoy wire